

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI.

COLLEGE OF SCIENCE

FACULTY OF PHYSICAL AND COMPUTATIONAL SCIENCES

BSc. COMPUTER SCIENCE

**CSM 478
COMPUTER NETWORKS**

STUDY GUIDE

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A. INTRODUCTION

Computer Networking is one of the fastest growing technologies today as almost every business depends on it for efficient operation. One of the consequences of this growth is the increasing demands for networking administrators who understands how computer networks operate. This course introduces students to computer networking concepts, with emphasis on Network Architectures, Network Design and Network Protocols. In addition, the course equips students with the knowledge and skills that can be applied towards preparation for an entry level professional networking certification such as CCNA for those interested in a computer networking career. The course is presented from a hands-on practical oriented approach to learning computer networking. Topics includes Types of Networks (LAN, WAN, etc), LAN Characteristics, LAN Topologies, WAN Characteristics, Switching, Bridging, Layered Network Architectures (OSI model, TCP/IP model), Addressing (MAC ADDRESSING, PORT ADDRESSING, IP ADDRESSING), DNS, Network Cabling, Packets Forwarding, Routing, Internetworking Devices, Subnetting, Socket Programming and Client-Server computing, Network Design and Implementation using CISCO Packet Tracer. In this course the four-layers Internet model are presented as the framework for the understanding of most current networking theories. Each layer builds upon the layer beneath it and supports the layer above it. In the same way, each concept introduced in this course builds upon the concepts examined in the previous sections. The Internet model was chosen because it is a protocol that is fully implemented. This course is designed for students with background in data communications. Review questions are used to test student's first level understanding of the material presented. Exercises requiring deeper participation of students are designed to create motivation for further study.

KEY TEXTBOOKS:

- **Hands-On Networking Fundamentals by Michael Palmer, Cengage Learning**

OTHER TEXTBOOKS:

- **CISCO Self Study: CCNA Study Guide (640-801) by Todd Lammle**
- **Data Communications and Networking by Behrouz A. Forouzan, 4th Edition**

B. COURSE OVERVIEW

The course is divided into six units. The first unit is an overview; the last unit is concerned with the Subnetting IP Addresses. The middle four units are designed to give more details of the computer networking concepts and design. The following summarizes the contents of each unit.

Unit One: Overview

The first unit gives a general overview of computer networking and its relationship to data communications. Chapter 1 of the textbook (Data Communications and Networking by Behrouz A. Forouzan, 4th Edition) covers the introductory concepts.

Unit Two: LAN and WAN Characteristics

The second unit describes types of networks (LAN and WAN) and their characteristics. Chapters 1 and 2 of the Key Textbook covers the contents presented in this unit.

Unit Three: Internetworking Devices and Function(s)

The third unit is devoted to the discussion of Network Devices and Devices Function(s) in Network. Chapter 5 of the key textbook as well as Chapters 1 of the textbook (CISCO Self Study: CCNA Study Guide (640-801) by Todd Lammle) covers this. In addition, Chapters 3 to 6 of the textbook (Data Communications and Networking by Behrouz A. Forouzan, 4th Edition) discusses telecommunication aspects of the physical layer. Chapter 7 introduces the transmission media and Chapter 8 is devoted to switching.

Unit Four: Network Models

The fourth unit describes the OSI and TCP/IP layered architectures that describes the interactivity between host on a computer network. Chapter 2 of the key textbook as well as Chapter 2 of the Textbook (Data Communications and Networking by Behrouz A. Forouzan, 4th Edition) covers the contents presented in this unit.

Unit Five: Internet Protocol (IP) Addressing

The five unit is devoted to the discussion of the Network Layer Logical Addressing (IP addressing) of the Internet model. Chapters 3 of the Key Textbook, Chapters 3 of the textbook (CISCO Self Study: CCNA Study Guide (640-801) by Todd Lammle) as well as Chapter 19 of the Textbook (Data Communications and Networking by Behrouz A. Forouzan, 4th Edition) covers the contents presented in this unit.

Unit Six: Subnetting IPv4 Addresses

The sixth unit is devoted to the discussion of the concept of Subnetting IPv4 addresses. Chapters 3 of the Key Textbook, Chapters 3 of the textbook (CISCO Self Study: CCNA Study Guide (640-801) by Todd Lammle) as well as Chapter 19 of the Textbook (Data Communications and Networking by Behrouz A. Forouzan, 4th Edition) covers the contents presented in this unit.

C. LEARNING TOOLS WITHIN THE COURSE

As you work through each unit, you will find these helpful tools and features:

- **Video Learning Sessions (VLS):** Recorded publications of live online teaching sessions is available to students and the text includes reminders about them. Depending on your personal learning style, you might use the videos in various ways. For example, one approach might be to review the previous lessons by watching the VLS.
- **Case in Point** This exciting feature provides mini cases for students to analyze and apply the skills and concepts learned.
- **Toolkit Time:** the CISCO Packet tracer will be used to model and simulate networks.
- **On the Web:** Learn more about a topic by visiting the suggested Web sites and exploring the links provided.

END OF UNIT EXERCISES

The following exercises are in every unit learned:

- **Learn It Online** Each unit features a Learn It Online page that includes six exercises. These exercises utilize the Web to offer unit-related reinforcement activities that will help you gain confidence in networks analysis and design. These exercises include True/False, Multiple Choice, and Short Answer questions.
- **Unit Exercises** These includes Review Questions and Discussion Topics. These exercises allow students to apply their understanding of the material and will help to prepare them for tests and assessments.
- **Apply Your Knowledge** This section includes use of mini cases that requires students to apply the knowledge and skills learned to real enterprise network scenarios.
- **Case Studies** Case studies provide practical experience and allow students to practice specific skills learned in the unit. Students completes their assignments using the CISCO Packet Tracer network modeling and simulation software.

D. COURSE OBJECTIVES:

On completion of this course, a learner should be able to.

- Understand the fundamental concepts of computer networking and its relationship with data communications.
- Describe the various types of computer networks (LAN, WAN, etc.)

- Demonstrate knowledge of LAN characteristics including Topologies, Transmission Modes, Transmission Methods, Access Methods, and Protocols
- Identify the different networking devices (Hub, Bridge, Switch, Router, & Gateway)
- Explain the function(s) of each networking device in a computer network
- Demonstrate knowledge of Collision Domain(s) and Broadcast Domain(s)
- Demonstrate knowledge of different Network Cabling and Interfaces
- Demonstrate knowledge of the layers of the Open Standard Interconnect (OSI) model
- Demonstrate knowledge of the functions of the layers of the TCP/IP model
- Demonstrate knowledge of IP Addressing and how IPv4 Addressing works
- Demonstrate knowledge of the concept of Subnetting and VLSM
- Apply the knowledge gained to design an enterprise computer network
- Implement the network design using a computer network simulation software such as CISCO Packet Tracer.

LEARNING OUTCOMES

*On completion of the **Computer Networks** course the students should be able to:*

- Please refer to course objectives above.

Teaching Approach:

Instructions will take the form of lectures, tutorials, problem solving, case studies and lab sessions with regular assignments.

Assessment:

Mid-semester exam: 15%
 Quiz/Assignments/Lab Sessions: 15%
 Final Examination: 70%

Recommended Reding Links:

CISCO Packet Tracer Tutorials

<https://www.computer-pdf.com/tutorials-introduction-to-cisco-packet-tracer>

CCNA LAB MANUAL

<https://sjce.ac.in/wp-content/uploads/2018/01/CCNA-lab-Manual.pdf>

ASSIGNMENT SCHEDULE

All assignments are due before the end of the specified day of delivery (GMT 23:59). All assignments are to be uploaded to the hand-in folder for this course unless other instructions are given. If you are unable to hand in your assignment on the LMS of KNUST (*vclass*), you may email it to the course facilitator (on the said day of delivery). Failure to deliver assignments on the specified date will attract penalties in the form of a reduced grade.

Assignment (Each Unit has an Assignment)	Description (Assignment title)	Type (Individual/Group)	Deadline (Duration)	Value (of final grade)
Explain and illustrate what Computer Network is? And discuss how computer networking relates to data communication	Introductory concepts	Individual	1 Week	
Explain all the different network types and outline their characteristics and functionalities?	Types of Computers Networks and Topologies	Individual	1 Week	
Explain how LAN and WAN communications works?	Network Models. TCP/IP and OSI Model	Individual	1 Week	
Evaluate the processes of Internet and Network Communications Protocols	Internetworking Concepts	Individual	1 week	
Create an enterprise networking model with Internetworking Devices	Internetworking configuration and management	Group	1 week	
Simulate an enterprise network model using CISCO Packet Tracer modeling and simulations Software	Enterprise Network Setup	Group	1 week	

* Participation in Online discussions may account for 10% of final grade for assessment

* Deadline could be weekly based

WEEKLY SCHEME OF LECTURES		
Week	Topic	Key Text Chapter(s)
1	Introductory Concepts - Fundamental concepts of computer networking and its relationship with data communications. ➤ Introduction to Networking ➤ What is Computer Network ➤ Components of Data Communication ➤ Benefits of Computer Networks ➤ Risks of Network Computing ➤ Types of Line Configuration ➤ Metric for assessing network effectiveness ➤ Data Representation and Number Systems	Chapters 1 Data Communications and Networking by Behrouz A. Forouzan
2 & 3	Introductory Concepts (ctnue) Describe the various types of computer networks (LAN, WAN, etc.) ➤ Local Area Networking (LAN) Communication ➤ Wide Area Networking (WAN) Communication ➤ Enterprise Network ➤ Communications Media	Chapters 1 & 2 Hands-On Networking Fundamentals by Michael Palmer
3	LAN characteristics including: ➤ LAN Topologies ➤ LAN Transmission Modes ➤ LAN Transmission Methods ➤ LAN Access Methods ➤ LAN Protocols	Chapters 1 & 2 Hands-On Networking Fundamentals by Michael Palmer
4 & 5	Network Devices and Devices Function(s) in Network Network Devices ➤ Network Interface Card (NIC) ➤ Hub ➤ Switch ➤ Bridge ➤ Ruters ➤ Gateway - Switching and Switching Tables ➤ Collision Domain(s) ➤ Broadcast Domain(s)	Chapters 1 CISCO Self Study: CCNA Study Guide (640-801)
6	Network Models The OSI Reference Model ➤ Physical Layer ➤ Data Link Layer ➤ Network Layer ➤ Transport Layer ➤ Session Layer ➤ Presentation Layer	Chapters 2 CISCO Self Study: CCNA Study Guide (640-801) Chapters 2 Data

	➤ Application Layer • The TCP/IP Protocol Suite ➤ Network Access Layer ➤ Internet Layer ➤ Host-to-Host Transport Layer ➤ Application Layer	Communications and Networking by Behrouz A. Forouzan Chapters 2 Hands-On Networking Fundamentals by Michael Palmer
7	Mid- Semester Exams	N/A
8	IP Addressing • Types of IP Addresses ➤ IPv4, IPv6 ➤ IPv4 Address Classification ➤ IPv4 Address Ranges ➤ Masking ○ Boundary Level/Fixed Length Mask ➤ IPv4 Reserved Addresses ○ Private/Non-Routable Addresses ○ Special Addresses ➤ IPv4 Public/Routable Addresses ➤ Network Prefix & Host Prefix • Network Map • Routing and Routing Tables • IPv6 Addresses LINK FOR STUDY MATERIAL IP Addressing and Subnetting for New Users https://www.cisco.com/c/en/us/support/docs/ip/routing-information-protocol-rip/13788-3.html Understanding IP Addressing: Everything you ever wanted to know https://pages.di.unipi.it/ricci/501302.pdf	Chapters 3 Hands-On Networking Fundamentals by Michael Palmer Chapters 3 CISCO Self Study: CCNA Study Guide (640-801) Chapters 19 Data Communications and Networking by Behrouz A. Forouzan
9	Subnetting • What is Subnetting? • Why Subnetting? • Subnetting IPv4 Classes A, B, & C IP Address • Non-Boundary Level Mask • Hierarchical IPv4 Addressing Scheme • Classless Inter-Domain Routing (CIDR) notation and impact on on IP Network Routing Tables • Variable Length Subnet Mask (VLSM) • Network Wiring Diagram using CISCO Packet Tracer Simulation Software LINK FOR STUDY MATERIAL IP Addressing and Subnetting for New Users	Chapters 3 Hands-On Networking Fundamentals by Michael Palmer Chapters 3 CISCO Self Study: CCNA Study Guide (640-801)

	https://www.cisco.com/c/en/us/support/docs/ip/routing-information-protocol-rip/13788-3.html Understanding IP Addressing: Everything you ever wanted to know https://pages.di.unipi.it/ricci/501302.pdf Download and Install CISCO Packet Tracer https://www.filehorse.com/download-cisco-packet-tracer-64/	Chapters 19 Data Communications and Networking by Behrouz A. Forouzan
10	Variable Lengths Subnet Mask (VLSM) • What is Variable Length Subnet Mask (VLSM) • How VLSM works • Network Wiring Diagram • Network Wiring Diagram using CISCO Packet Tracer Simulation Software LINK FOR STUDY MATERIAL IP Addressing and Subnetting for New Users https://www.cisco.com/c/en/us/support/docs/ip/routing-information-protocol-rip/13788-3.html Understanding IP Addressing: Everything you ever wanted to know https://pages.di.unipi.it/ricci/501302.pdf Download and Install CISCO Packet Tracer https://www.filehorse.com/download-cisco-packet-tracer-64/	Chapters 3 CISCO Self Study: CCNA Study Guide (640-801)

Teaching Approach:

Instructions will take the form of lectures, tutorials, problem solving, case studies and lab sessions with regular assignments.

Assignments:

Please visit the vclass room for all assignments.

END-OF- COURSE EVALUATION

Please visit the End of Course evaluation folder to pick the evaluation questionnaire. Answer the questions and submit your response to the facilitator. This evaluation is very critical to the betterment of the course in subsequent sessions.

The final examination, which counts towards 60% of your final grade, will include all topical issues discussed during this course. Please review all groups' reports, debates and individual assignments, as preparation for your final exam. The programme's examination officer will communicate the final examination dates and venue to students sometime later.
 Thanks for your cooperation!!!☺

END-OF-SEMESTER EVALUATION – CSM 478 COMPUTER NETWORKS

sn	Issues	1	2	3	4	5
Factor 1: Appropriateness of Readings and Assignments						
	Assignments were appropriate and effective for learning course content.					
	Discussion board assignments fostered a high level of interaction among students.					
	Online assignments included authentic, real-life activities.					
	Discussion board activities were designed to evoke further critical thinking about course content.					
	Selected readings and resources were adequate for the course objectives.					
Factor 2: Technological Tools						
	Links were descriptive and provided information regarding the content.					
	Links to external readings opened in new windows.					
	Interactive multimedia items allowed students to control content.					
	Technological requirements to complete the online course were specified.					
	Technological tools were used appropriately for the course content					
	Online course content included varied types of assignments to appeal to different learning Style					
Factor 3: Instructor Feedback and Communication						
	Feedback was informative and clearly articulated.					
	Feedback was delivered in a timely manner.					
	Instructor effectively communicated any changes/clarified any misunderstanding regarding course requirements.					
Factor 4: Course Organization						
	The sequence of online course activities was effectively organized and easy to follow					
	Dates on the syllabus and course schedule corresponded to online readings					
	Dates on the course schedule corresponded to drop box and discussion board submissions					
	Online course content was consistent in presentation.					
Factor 5: Clarity of Outcomes and Requirements						
	Assignments and activities were clearly linked to the course objective.					
	Evaluation criteria were clearly stated					
	Evaluation criteria for discussion board activities were clearly specified in advance					
	Requirements for drop box submissions were specified and easy to follow.					
	The course has evoked further interest in this field					
Factor 6: Content Format						
	Online course materials were free of spelling errors and grammatical errors					
	Font size and layout of the online content was consistent					

The scale consisted of 25 likert type items, using a five-point scale (1 = strongly disagree; 2 = disagree, 3 = neutral; 4 = agree; 5 = strongly agree). **Source: Rothman, T., Romeo, L., Brennan, M., & Mitchell, D. (2010). 21st century best practice and evaluation for online courses. In International Conference: The Future of Online Education**